

Remote Sensing Image Change Caption Generation

Evidence Report — Predictions, Curves, Maps and Plots

Independent Study, Monsoon 2026

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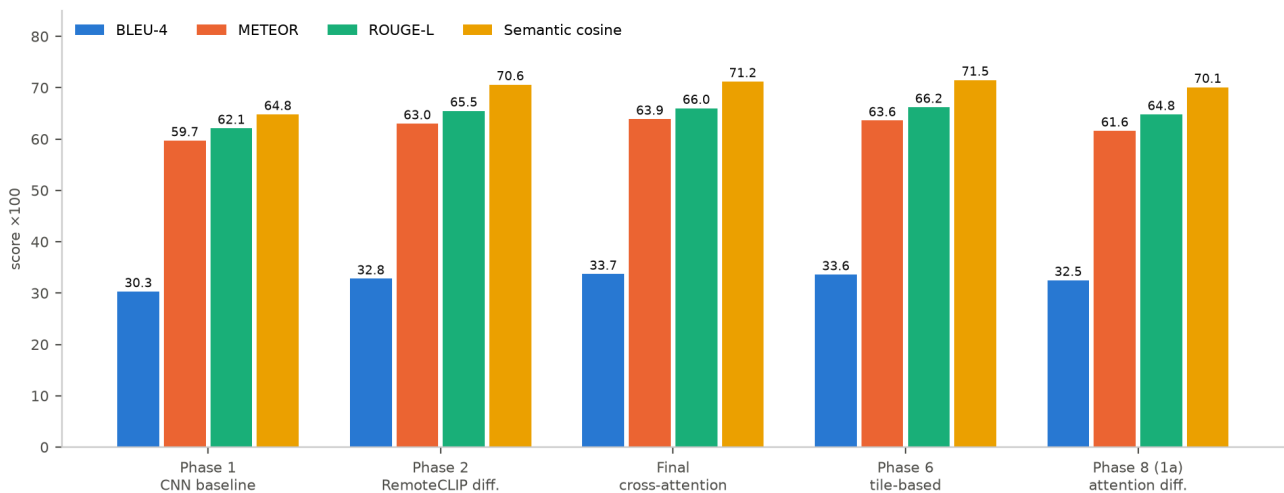
This report collects the evidence behind the observations in the Main Report. Each section names the observation it supports (Obs. 1–15).

How to read this report

- **Sources.**
 - Captioning results come from the notebook outputs of the [project repository](#).
 - Prediction samples come from the saved prediction sheets, published as [RSICC Model Run Reports](#) on Kaggle.
 - The Indian-dataset evidence comes from the experiment logs and the labelled files, published as [India Sentinel-2 Change Pairs](#) and its [labelled subset](#).
- **Ground truth for Part C** is the human label where one exists.
- **Prediction-sheet naming.** The sheets named `predictions_phase7*` contain Phase 6 model outputs. They are labelled "Phase 6 run" here.

Part A — Captioning models on LEVIR-CC and SECOND-CC

E-A1. Scores per model on LEVIR-CC test (Obs. 1, 2)

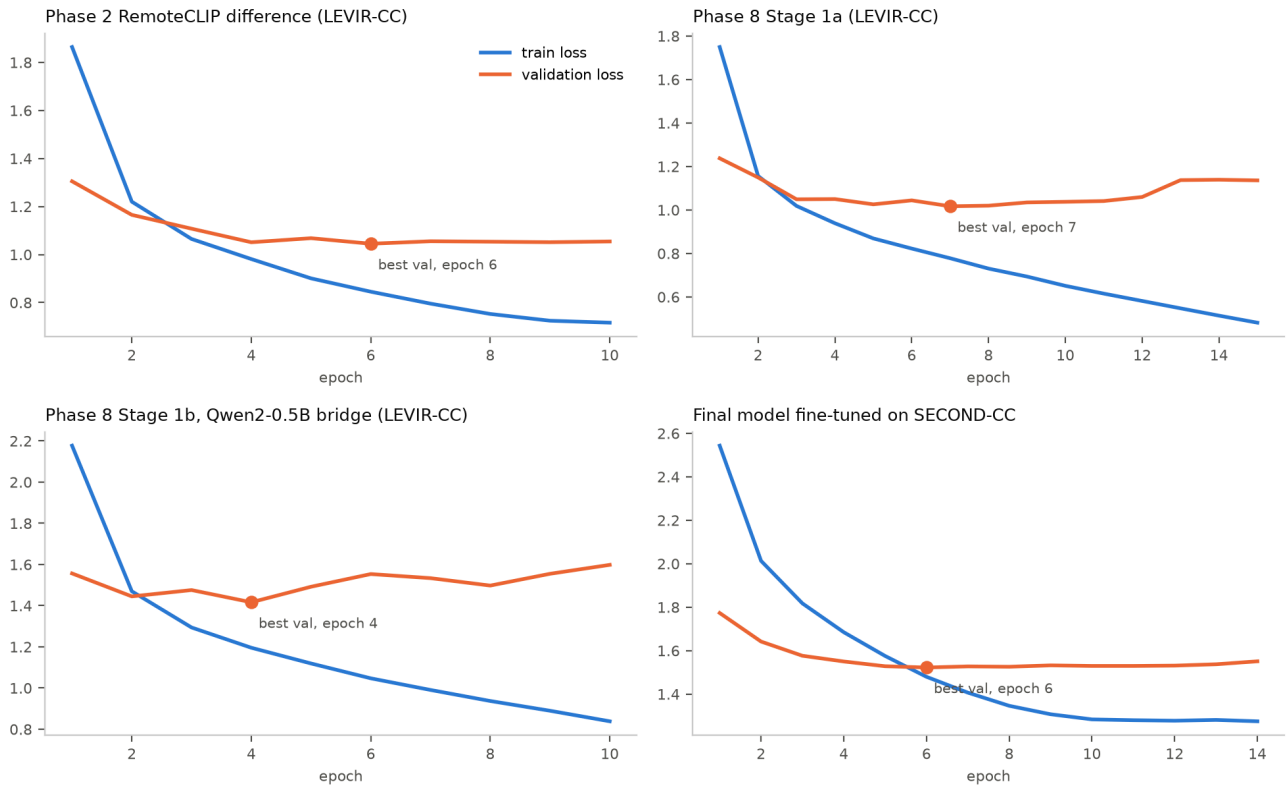


Model	Loss	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	Cosine
Phase 1 baseline	0.9690	0.4720	0.3855	0.3393	0.3027	0.5966	0.6214	0.6477
Phase 2 RemoteCLIP difference	0.9345	0.5168	0.4237	0.3712	0.3284	0.6304	0.6545	0.7061
Final cross-attention	0.8982	0.5311	0.4355	0.3810	0.3365	0.6392	0.6604	0.7123
Phase 6 tiles	0.9225	0.5214	0.4297	0.3777	0.3358	0.6357	0.6624	0.7149
Phase 8 stage 1a (epoch 7)	—	0.5092	0.4167	0.3658	0.3250	0.6155	0.6482	0.7007
Phase 8 stage 1b (Qwen bridge)	—	0	0	0	0	0	0	0.0104

Phase 1 to Phase 2 (encoder only changed, 3.41 M vs 3.43 M trainable parameters):

- BLEU-1 to BLEU-4: +8.5% to +9.9% relative
- METEOR +5.7%, ROUGE-L +5.3%, cosine +9.0%
- test loss −3.6%

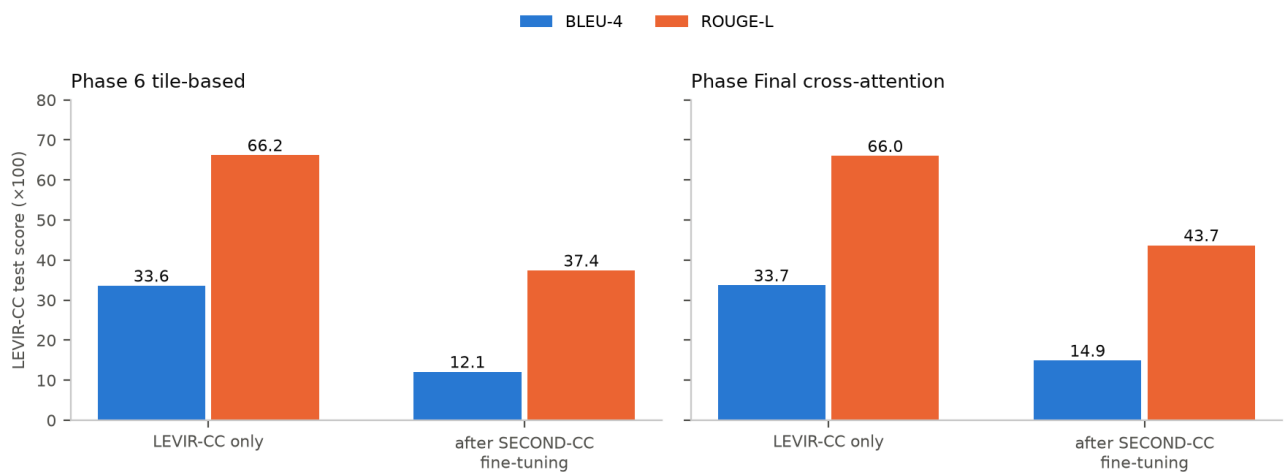
E-A2. Training and validation curves (Obs. 3, 6)



In every run, validation loss reaches its minimum early while training loss keeps falling:

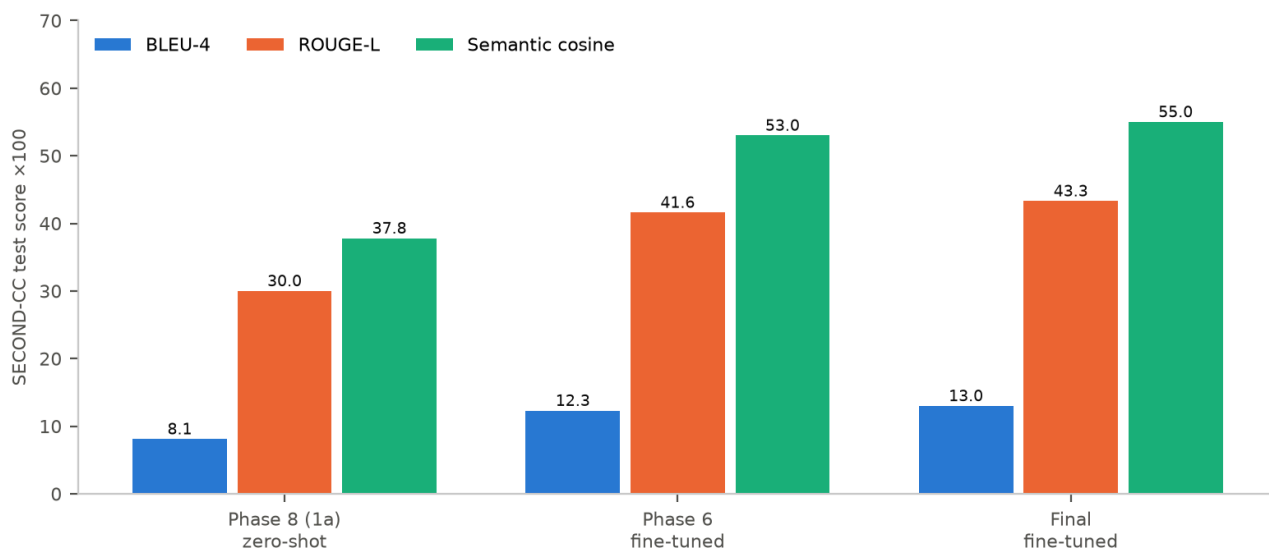
Run	Best validation loss	Final validation loss	Final training loss
Phase 2	1.045 (epoch 6)	1.054 (epoch 10)	0.716
Phase 8, stage 1a	1.017 (epoch 7)	1.136 (epoch 15)	0.482 (training accuracy 84.9%)
Phase 8, stage 1b	1.417 (epoch 4)	1.598 (epoch 10)	—
SECOND-CC fine-tuning	1.522 (epoch 6)	1.551 (epoch 14)	—

E-A3. Forgetting after SECOND-CC fine-tuning (Obs. 6)



Model	LEVIR-CC loss	BLEU-4	METEOR	ROUGE-L	Cosine
Final, before fine-tuning	0.898	0.337	0.639	0.660	0.712
Final, after fine-tuning	4.080	0.149	0.394	0.437	0.458
Phase 6, before fine-tuning	0.923	0.336	0.636	0.662	0.715
Phase 6, after fine-tuning	3.520	0.121	0.318	0.374	0.438

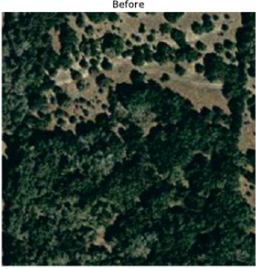
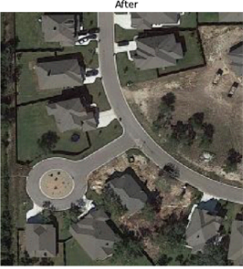
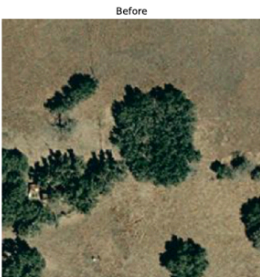
E-A4. Domain shift to SECOND-CC (Obs. 5)



Model / condition	BLEU-1	BLEU-4	METEOR	ROUGE-L	Cosine
Phase 8 stage 1a, zero-shot	0.179	0.081	0.298	0.300	0.378
Phase 6, fine-tuned	0.354	0.123	0.392	0.416	0.530
Final, fine-tuned	0.363	0.130	0.413	0.433	0.550

The zero-shot and fine-tuned rows come from different models, so this comparison is indicative.

E-A5. The same LEVIR-CC test pairs across three runs (Obs. 4, 7)

Before 	After 	LEVIR-CC test sample 4 Reference: the vegetation has been replaced by a road and many villas around . Phase 6 run: a huge residential area is built in almost whole scene Phase 8 run: the vegetation quantity is increased in many parts of the scene Final, after SECOND-CC fine-tuning: there is no difference
Before 	After 	LEVIR-CC test sample 17 Reference: a road appears at the bottom and many houses are scattered replacing the trees . Phase 6 run: a huge residential area is built in almost whole scene Phase 8 run: many buildings are constructed in the middle of the scene Final, after SECOND-CC fine-tuning: there is no difference
Before 	After 	LEVIR-CC test sample 19 Reference: the trees have been replaced by many buildings . Phase 6 run: a huge residential area is built in almost whole scene Phase 8 run: many buildings are constructed in the middle of the scene Final, after SECOND-CC fine-tuning: there is no difference
Before 	After 	LEVIR-CC test sample 42 Reference: a house appears at the top right corner . Phase 6 run: there is no difference Phase 8 run: there is no difference Final, after SECOND-CC fine-tuning: there is no difference
Before 	After 	LEVIR-CC test sample 2 Reference: there is no difference . Phase 6 run: the vegetation quantity is decreased in many parts of the scene Phase 8 run: there is no difference Final, after SECOND-CC fine-tuning: there is no difference

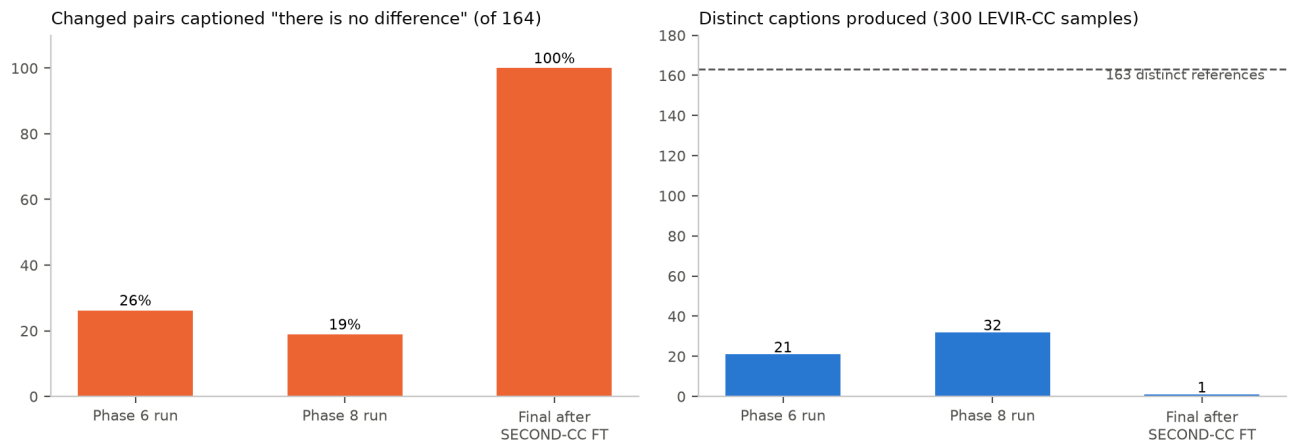
Sample 4, 17 and 19 (forest or scrub replaced by a housing estate):

- The Phase 6 run gives a template ("a huge residential area is built...").
- The Phase 8 run partly contradicts the scene ("vegetation quantity is increased").
- The Final model after fine-tuning says "there is no difference".

Sample 42 (one house added at the top right): all three runs miss it.

Sample 2 (no change): the Phase 6 run reports vegetation loss.

E-A6. Prediction behaviour across 300 saved LEVIR-CC samples (Obs. 4, 7)



Prediction sheet	Changed pairs called "no difference"	Unchanged pairs correct	Distinct captions
Phase 6 run	43 / 164	98 / 136	21
Phase 8 run	31 / 164	96 / 136	32
Final after SECOND-CC fine-tuning	164 / 164	136 / 136	1
References	—	—	163

On the SECOND-CC prediction sheets (300 samples, 255 changed), the Phase 6 and Phase 8 runs produced 97 and 82 distinct captions. They missed 16 and 19 changed pairs, and called 12 and 15 of 45 unchanged pairs changed.

E-A7. SECOND-CC test predictions (Obs. 5)

SECOND-CC test samples (Phase 6 run)

Before

After



Sample: 1

Reference:
the trees is turned into bareland at the middle of the scene

Prediction:
the vegetation quantity is increased in the middle of the scene

Before

After



Sample: 5

Reference:
trees are removed

Prediction:
the vegetation quantity is increased in the middle of the scene

Before

After



Sample: 9

Reference:
lush area turned bareland in the center

Prediction:
the blue building at the corner of the scene is replaced with a dirt area

E-A8. Whole-image vs patch-wise inference on unlabelled high-resolution pairs (Obs. 9)

Whole-image inference



Patch-wise inference



These 43 pairs have no reference captions. In the samples shown, the after images contain new farm structures and buildings.

- **Whole-image inference** (23 distinct captions) mostly reports "vegetation quantity is decreased".
- **Patch-wise inference** (9 distinct captions) gives "the green region is replaced by a huge house at the bottom right" for 75% of pairs, regardless of where the change is.

Part B — Off-the-shelf captioners on Indian imagery

E-B1. Pilot Indian Sentinel-2 collection



60 areas of interest in 5 categories, tiled into 512×512 patches and filtered by change magnitude. Of 2,126 tiles checked, 1,072 pairs were kept, in 1 h 53 min with 12 workers.

E-B2. Captioner outputs (Obs. 10)



Before

After

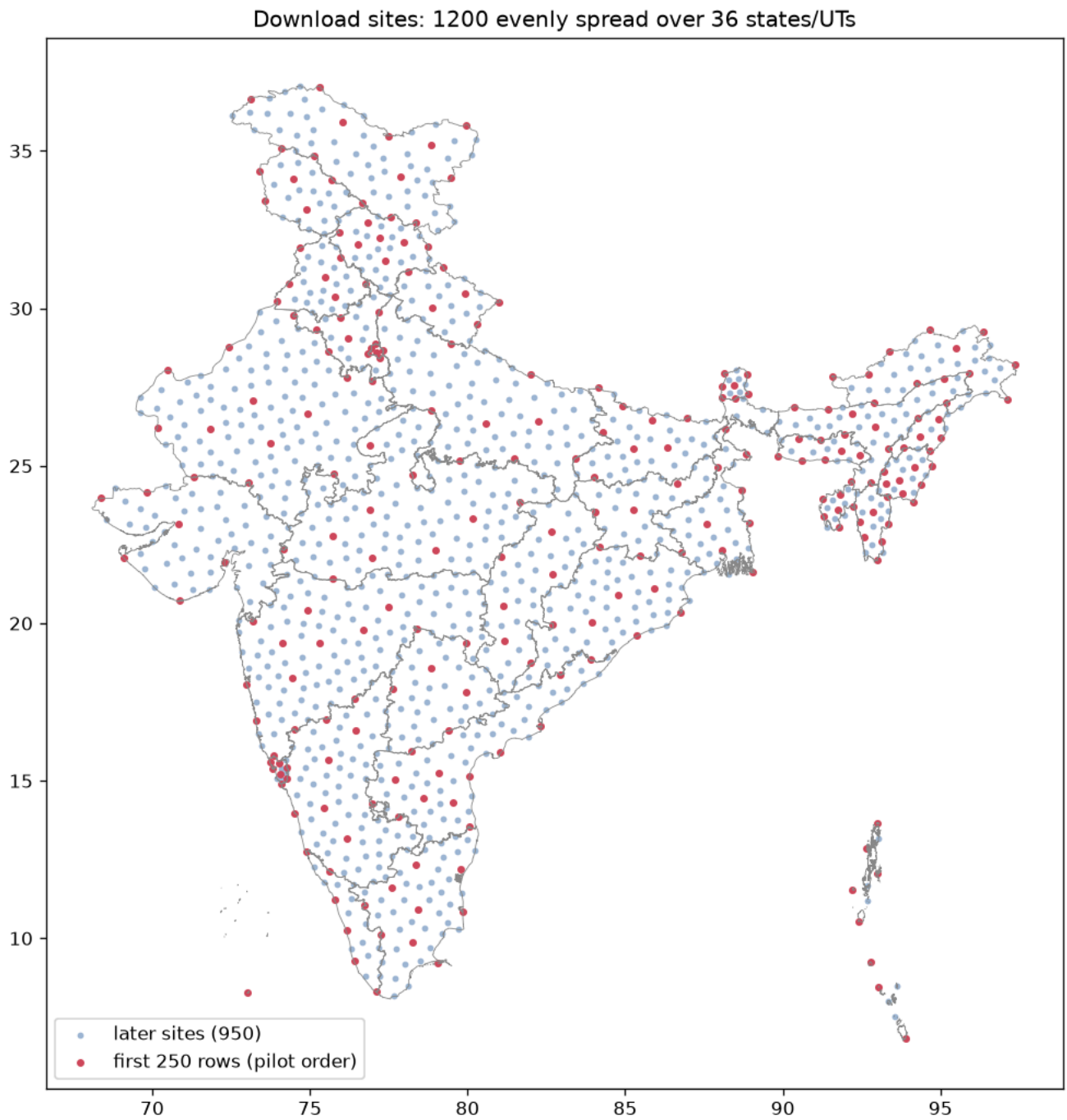
Qwen2.5-VL: "...increased urban density and expansion of buildings, particularly along the central axis, indicating rapid development. There's also visible growth in road infrastructure..."

TeoChat: "Nothing meaningful changed."

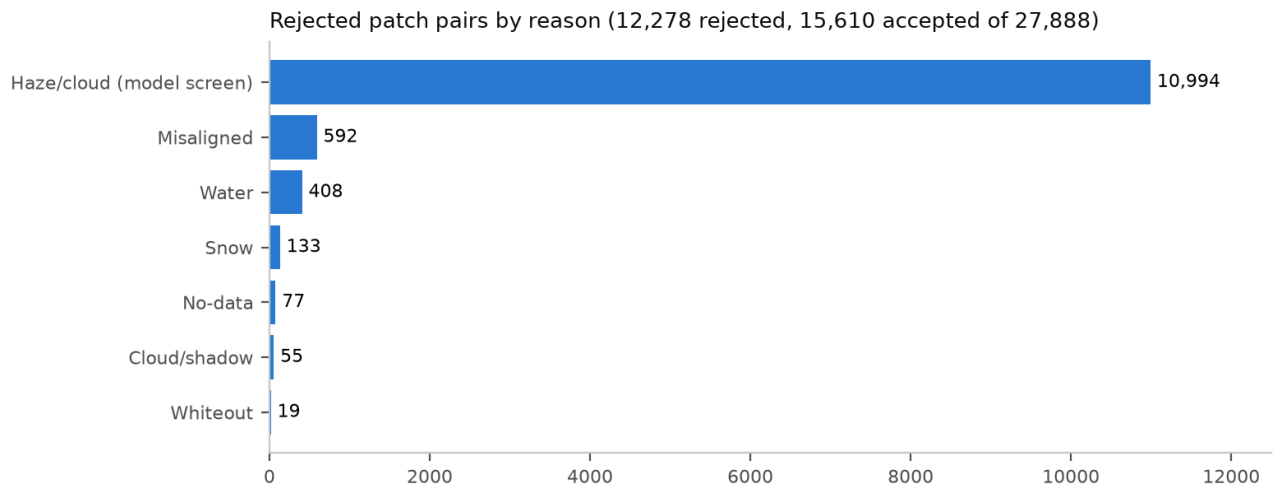
Captioner (30 pairs)	Outcome
TeoChat	"Nothing meaningful changed" on 27/30; one invented hurricane-damage description
TeoChat, 4-bit	"Nothing meaningful changed" on 23/30
Qwen2.5-VL, first run	11 empty and 12 garbled answers (non-English text, repeated tokens)
Qwen2.5-VL, retries / 8-bit	Failures reduced but not removed; answers describe generic urban development

Part C — India-wide dataset and automatic labelling

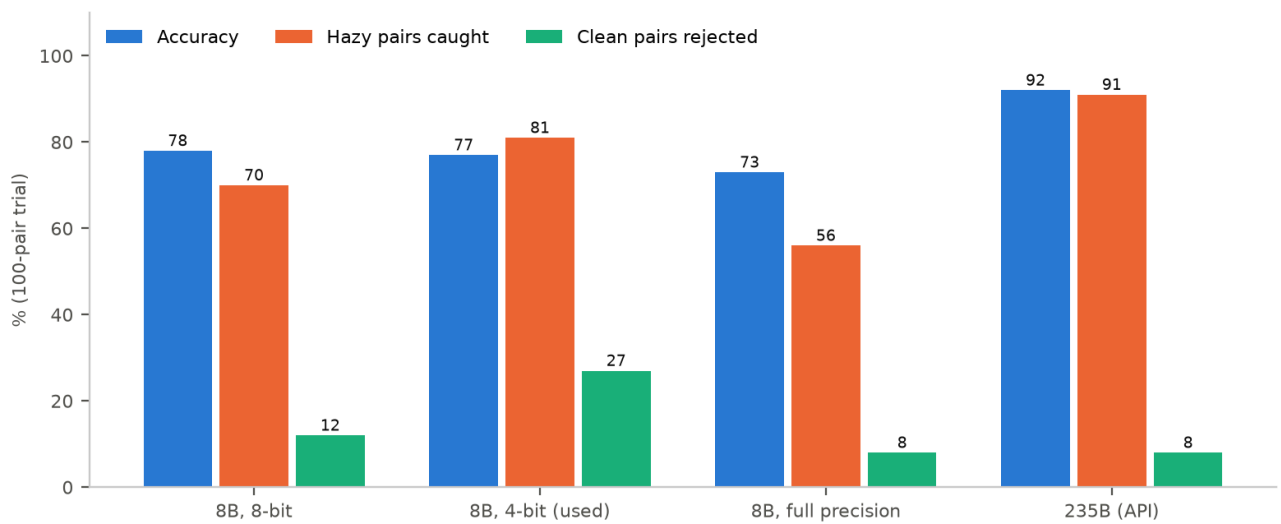
E-C1. Site coverage



E-C2. Screening outcome (Obs. 11)



E-C3. Haze screen trial, 100 pairs (Obs. 11)



The 4-bit model was used for the full run because it was 8× faster on the cluster GPU (1.4 s per pair). The cost was a 27% wrong-reject rate on clean pairs. The reference labels for this trial were re-checked at large size using the 235B model as a second opinion, which favours the 235B row; its accuracy should be read as an upper bound.

E-C4. Label classes and human review (Obs. 12)

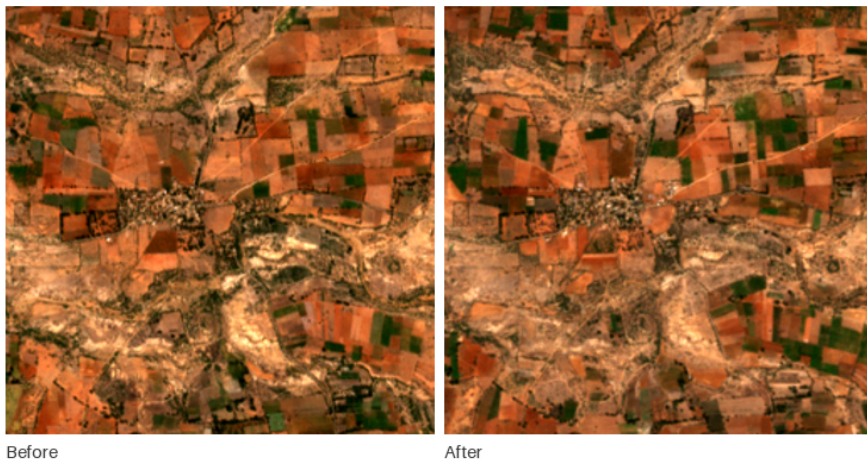
Human label: Change (andhra_pradesh_t1_024_r1c1)



Human label: Seasonal (andhra_pradesh_001_r0c2)



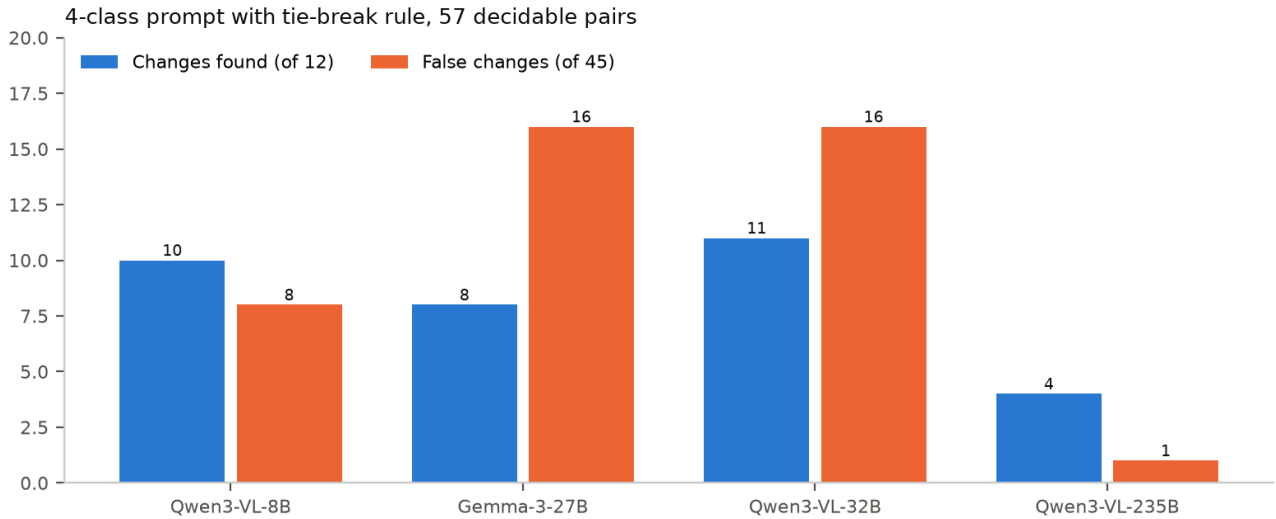
Human label: No change (andhra_pradesh_001_r1c2)



	Change	Seasonal	No change	Can't tell	Total
Human-labelled (blind)	55	214	226	40	535
All labelled pairs	143	400	383	76	1,002

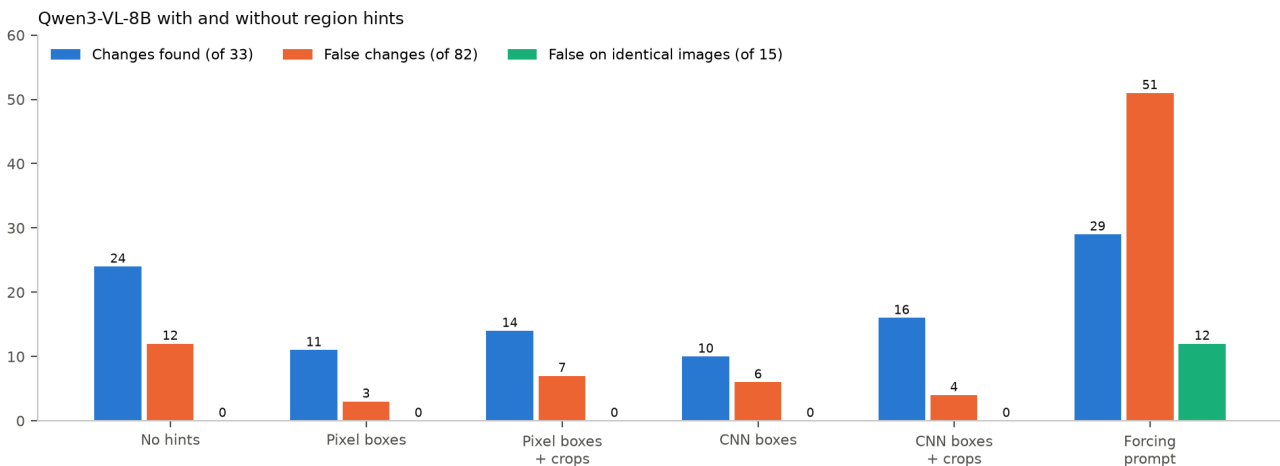
- The human agreed with the automatic suggestion on 84% of reviewed pairs.
- In a random sample of 45 pairs where everything agreed, the human disagreed on 9: 4 changes everyone missed, and 5 false changes everyone accepted.

E-C5. Model size comparison (Obs. 13)



Asked plainly, the 235B dismissed real construction as "seasonal or agricultural variation" in most missed cases. The 32B over-called change.

E-C6. Effect of region hints and forcing prompts (Obs. 14)



The no-hint baseline used the 4-class prompt, while the boxed variants used a per-box prompt; the box variants are directly comparable with each other. With region boxes, the model's overall answer matched its answer for the boxes. It never reported a change outside the boxes, even when asked to.

E-C7. Example: region boxes hide a real change (Obs. 14)

Plain prompt: change found (correct). Human label: Change (bihar_004_r2c1)



Before



After

Pixel boxes: answer 'none' (missed)

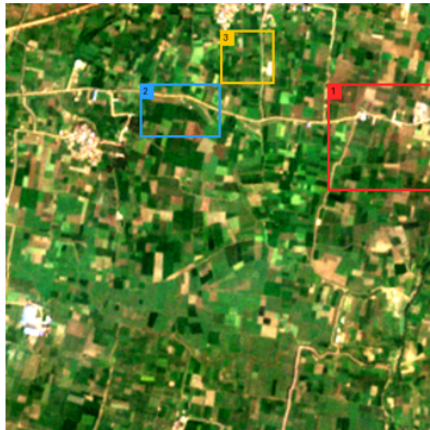


Before, pixel-difference boxes



After, pixel-difference boxes

CNN boxes: answer 'seasonal' (missed)



Before, CNN boxes

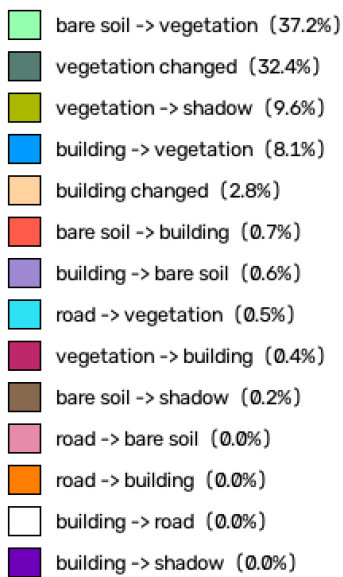


After, CNN boxes

E-C8. Rule-based change map (Obs. 14)

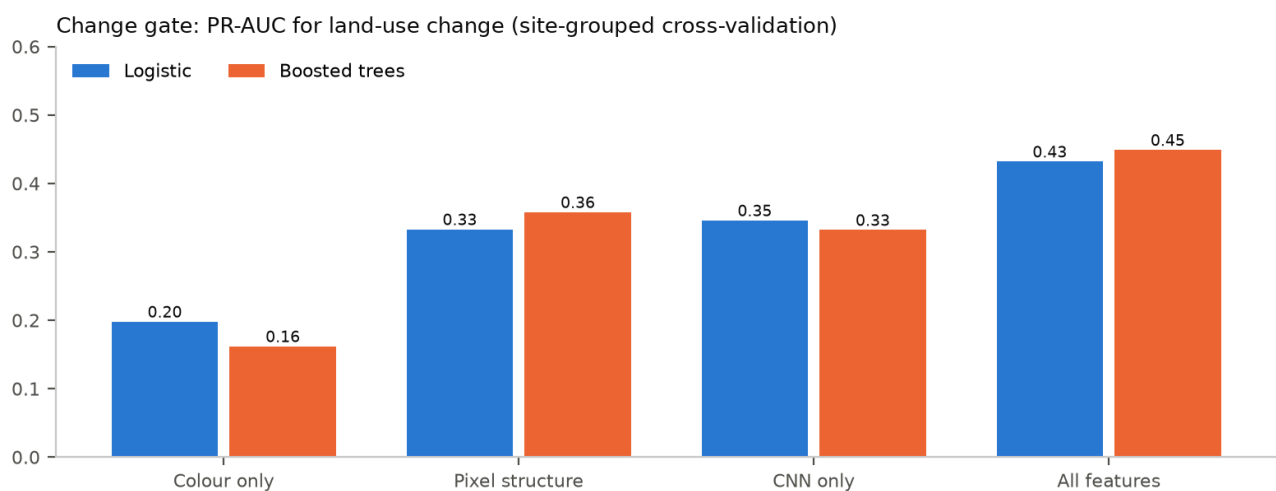
Human label: Seasonal (madhya_pradesh_071_r3c3). Before | After | change map

32B caption with map: "Large areas of bare soil became vegetated, vegetation changed across the scene, and buildings were removed in the left and bottom-left, leaving vegetation."



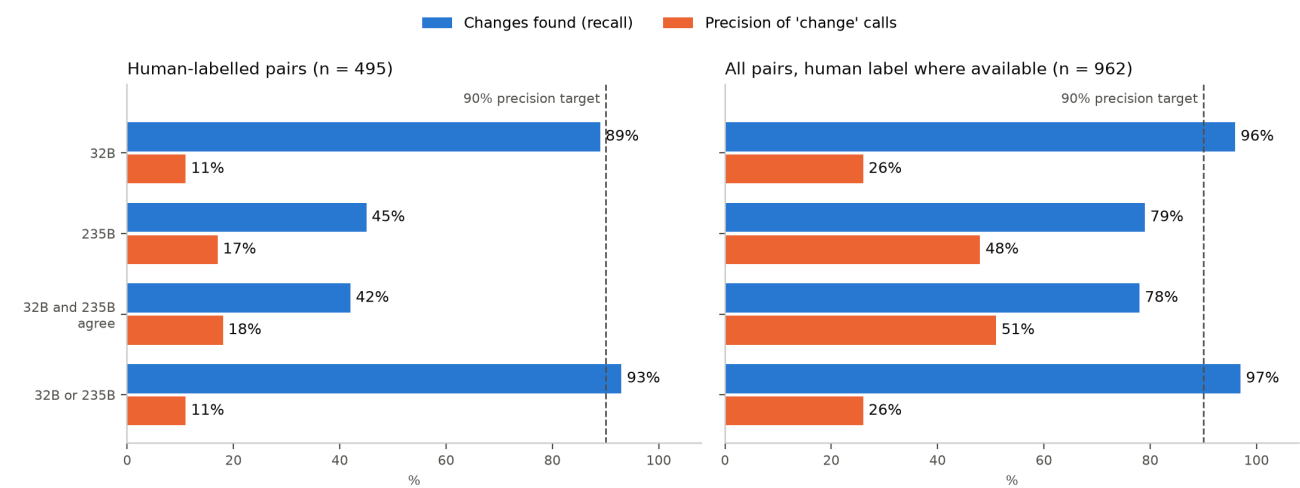
On 50 pairs, the map marked a median 82% of each image as changed (maximum 98%). With the map treated as fact, the model reported change on 19/19 real-change pairs and on 27/27 no-change pairs.

E-C9. Pixel and CNN feature change gate



At 90% recall the gate keeps about 62% of pairs, so it can only rank pairs, not label them.

E-C10. 1,000-pair validation (Obs. 15)



Human-labelled set (n = 495), with 95% Wilson intervals:

Rule	Changes found	Precision	False-change rate
32B	89% (78–95)	11% (9–15)	87% (84–90)
235B	45% (33–58)	17% (12–24)	27% (23–32)
Both agree	42% (30–55)	18% (12–25)	24% (21–29)

How often each model said "change", by human label:

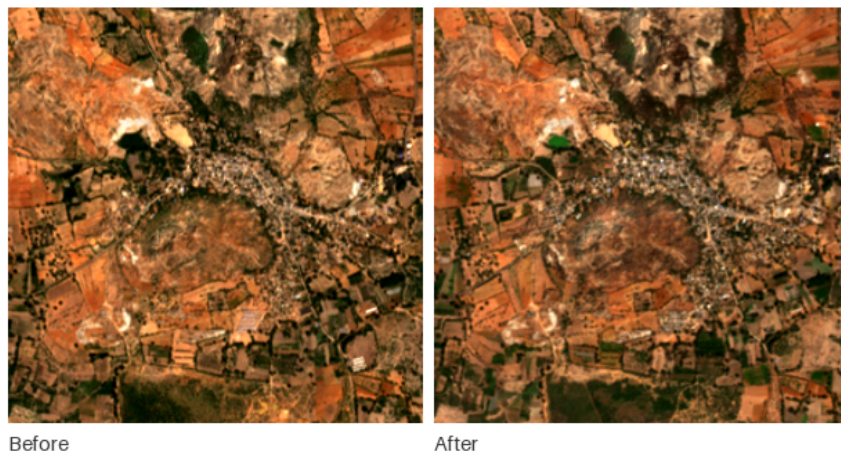
Human label	32B says change	235B says change
Seasonal (n = 214)	200 (93%)	67 (31%)
No change (n = 226)	184 (81%)	53 (23%)
Change (n = 55)	49 (89%)	25 (45%)

E-C11. Validation examples (Obs. 13, 15)

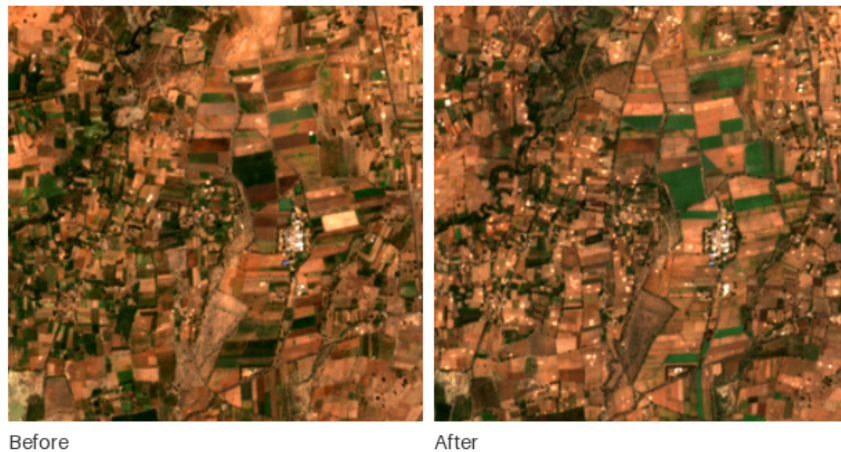
(a) Correct: human Change; 32B and 235B say change



(b) False alarm: human Seasonal; 32B and 235B say change



(c) Missed: human Change; 32B and 235B say seasonal



The 235B's conclusions in its own words:

- **(a) Correct, human label Change:** *"There has been significant, lasting development, including new buildings, roads, and land clearing."*
 - **(b) False alarm, human label Seasonal:** *"Lasting changes have occurred, including new agricultural plots, possible expansion of settlement."*
 - **(c) Missed, human label Change:** *"The differences are not lasting structural changes. They are consistent with seasonal or annual agricultural cycles."*
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Resources

Resource	Link
Code repository	github.com/Sankalp0109/Generative-and-Caption-Based-Vision-Language-Models-for-Remote-Sensing-Change-Detection
India Sentinel-2 Change Pairs (15,610 pairs)	kaggle.com/datasets/kspsvlnsiddardha/india-sentinel2-change-pairs
Labelled subset (1,002 pairs)	kaggle.com/datasets/kspsvlnsiddardha/india-sentinel2-change-labelled
Model run reports (prediction sheets)	kaggle.com/datasets/kspsvlnsiddardha/rsicc-model-run-reports
Model checkpoints	kaggle.com/models/kspsvlnsiddardha/rsicc-change-captioning · huggingface.co/Kspsvln/IS · huggingface.co/Kspsvln/IS_phase2

Datasets, models and metrics are cited in the References of the Main Report. Contains modified Copernicus Sentinel data (2019–2026).